

Project Summary

Team name	SENTRY	
Project title	Low-Cost Bridge Vibration Anomaly Detection System Robust to Temperature and Humidity Changes	
Role	Name	Affiliation
Participants	Kim Changhoon	CheongShim International Academy
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Project summary		
1. Project topic We developed a low-cost first-stage bridge vibration screening system that observes vibration and the environment together with a MEMS accelerometer and a temperature/humidity sensor, compensates for normal variation explained by temperature and humidity, detects only the remaining change as anomaly candidates, and also shows the bands that departed. 2. Description of the project (proposed solution and its rationale) Class 3 small and medium bridges, 64.5% of all bridges in Korea, lack a low-cost means of continuous monitoring to fill the gap between precision diagnoses. This project collects 500 Hz vibration and temperature/humidity with an ADXL355Z, a DHT22 and a Raspberry Pi 5 (about ₩190,000) under excitation from a solenoid auto-tapper, compensates eight log/ratio-normalized features with a temperature/humidity regression learned on normal data, trains an Autoencoder unsupervised on the residuals alone, flags an anomaly candidate when the reconstruction error exceeds the threshold, and shows the band contributions on a dashboard. The design rests on the facts that natural frequencies change with temperature and humidity even without damage and that damage data cannot be obtained from real bridges. 3. Differences from existing work (solutions) Existing low-cost MEMS studies do not address environmental compensation, KRISS's fiber-optic (BOCDA) precision measurement requires dedicated infrastructure, and unsupervised studies on public data mostly stop at outputting an anomaly score. This project is distinguished by integrating measurement, environmental compensation, anomaly judgment and band-wise explanation into a single low-cost device. 4. Results In the environment × mass × bolt experiment with 27 conditions (1,380 windows), the compensated model detected all 840 damage windows (recall 100%, F1 0.998), with 4/90 false alarms (4.4%). Over 5 repeated seeds, only the compensated model kept 100% for every seed (uncompensated 96.2-99.3%). The anomaly score grew with the severity of damage (5.8× → 22.6×), and bolt loosening (40-80 Hz) and mass change (amplitude/energy) were distinguished by their contributing features. The reduction in false alarms from compensation was confirmed in the humidity segment of the stage-1 experiment, 17.5% → 9.1%, and was reproduced in the same direction on full-scale openLAB data.		